Respiration and gas exchange

25 questions: 8 multiple choice, 17 structured.

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1 · Part 3

Paper 1 & 2 Respiration [1]

Which statement about ATP is correct?

A ATP is a type of sugar that cells store as an energy reserve.

B ATP is produced by respiration and provides energy for processes in cells.

C ATP is produced by aerobic respiration but not by anaerobic respiration.

D ATP is broken down during respiration to release glucose.

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2 · Part 3

Paper 1 & 2 Respiration [1]

Which row describes anaerobic respiration in human muscle cells?

	Is oxygen needed?	Energy released from each glucose molecule, compared with aerobic respiration	Is carbon dioxide produced?
A	yes	less	no
B	no	more	no
C	no	less	yes
D	no	less	no

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3 · Part 3 Paper 1 & 2 Respiration [1]

Germinating bean seeds were placed in a vacuum flask with a thermometer. The flask was kept in a dark cupboard. After three days, the temperature inside the flask had risen by 6 °C.

Why did the temperature inside the flask rise?

A Respiration in the seeds released some energy as heat.

B The seeds absorbed heat from the air in the cupboard.

C The seeds photosynthesised, and this released heat.

D Water evaporating from the seeds released heat.

4 · Part 3 Paper 1 & 2 Gas exchange [1]

Which substance in tobacco smoke combines with haemoglobin in red blood cells and reduces the amount of oxygen the blood can carry?

A carbon dioxide

B nicotine

C carbon monoxide

D tar

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5 · Part 3 Paper 1 & 2 Gas exchange [1]

Which row describes what happens when a person who is sitting still breathes out normally?

	Intercostal muscles	Diaphragm	Air pressure inside the lungs
A	relax	relaxes	increases
B	contract	contracts	increases
C	relax	contracts	decreases
D	contract	relaxes	decreases

6 · Part 3

Paper 2 only | Gas exchange | [1]

Carbon dioxide from the air enters a leaf. How does the carbon dioxide reach the palisade mesophyll cells inside the leaf?

A. It is carried to the cells in the xylem vessels.

B. It is moved into the cells by active transport from the leaf surface.

C. It moves into the cells by osmosis.

D. It diffuses through the air spaces inside the leaf and then into the cells.

7 · Part 3

Paper 2 only | Gas exchange | [1]

In bright light, there is a net movement of oxygen out of a leaf by diffusion. Why does this happen?

A. The leaf uses oxygen faster than it produces oxygen.

B. Oxygen is moved out of the leaf by active transport.

C. The oxygen concentration inside the leaf is higher than in the air outside.

D. The oxygen concentration in the air outside is higher than inside the leaf.

8 · Part 3

Paper 2 only | Gas exchange | [1]

Hydrogen-carbonate indicator is red when it is in equilibrium with normal air. It turns yellow when the carbon dioxide concentration increases and purple when the carbon dioxide concentration decreases.

Sealed tubes of red hydrogen-carbonate indicator were set up, some containing a leaf held above the indicator. The tubes were left for three hours at the same temperature.

Which row shows a correct result?

	Contents of tube	Conditions	Colour of indicator after three hours
A	leaf	bright light	yellow
B	leaf	dark	yellow
C	no leaf	bright light	purple
D	leaf	dark	purple

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9 · Part 3

Paper 1 & 2 Respiration [6]

Aerobic respiration takes place in the cells of all plants and animals.

(a) Give the word equation for aerobic respiration. [2]

(b) Give the balanced chemical symbol equation for aerobic respiration. [2]

(c) Respiration happens in all living cells all of the time. Explain why a cell dies if respiration stops. [2]

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10 · Part 3

Paper 1 & 2 Respiration [5]

Kenji ran on a treadmill at six different speeds. After four minutes at each speed, a small sample of his blood was taken and the concentration of lactic acid in the blood was measured.

The table shows the results.

Running speed / km per hour	Lactic acid concentration in blood / mmol per dm^3
8	1.0
10	1.1
12	1.4
14	2.5
16	4.8
18	8.6

(a) Lactic acid is produced by anaerobic respiration in muscle cells. Give one way in which anaerobic respiration is different from aerobic respiration, other than the substances produced. [1]

(b) Describe how the lactic acid concentration in Kenji's blood changes as his running speed increases. [2]

(c) Explain the change in lactic acid concentration at the higher running speeds. [2]

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11 · Part 3

Paper 1 & 2 Respiration [5]

After heavy rain, the spaces between soil particles can fill with water. The soil is then waterlogged, and the roots of plants growing in it get very little oxygen. The root cells then respire anaerobically.

(a) Give the word equation for anaerobic respiration in plant cells. [2]

(b) Root cells take up mineral ions from the soil by active transport.

Explain why a plant growing in waterlogged soil takes up mineral ions more slowly. [3]

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12 · Part 3

Paper 1 & 2 Respiration [7]

Priya investigated the heat released by germinating peas.

- She soaked some peas in water for one day so that they started to germinate.
- She boiled half of the peas for ten minutes and then let them cool to room temperature.
- She rinsed both lots of peas in a dilute disinfectant solution.
- She filled vacuum flask X with living germinating peas and vacuum flask Y with the boiled peas.
- She placed a thermometer in each flask and closed the top of each flask with a cotton wool plug.
- She recorded the temperature in each flask once a day for four days.

Time / days	Temperature in flask X / °C	Temperature in flask Y / °C
0	20.0	20.0
1	22.6	20.0
2	24.8	20.0
3	26.8	20.0
4	28.4	20.0

(a) Explain why Priya set up flask Y with boiled peas. [2]

(b) Explain why Priya rinsed both lots of peas in disinfectant. [2]

(c) Calculate the mean rate of temperature increase in flask X over the four days.

Use the formula: mean rate of temperature increase = (final temperature − starting temperature) ÷ number of days [2]

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(d) Explain why Priya used vacuum flasks and not open glass beakers. [1]

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13 · Part 3

Paper 1 & 2 Respiration [6]

Hydrogen-carbonate indicator is red when it is in equilibrium with normal air. It turns yellow when the carbon dioxide concentration increases and purple when the carbon dioxide concentration decreases.

Amara investigated the effect of temperature on respiration in woodlice.

- She put 10 cm^3 of red hydrogen-carbonate indicator in a boiling tube.
- She placed a gauze platform in the tube above the indicator and put five woodlice on the platform.
- She sealed the tube with a bung and placed it in a water bath at 15 °C.
- She recorded the time taken for the indicator to turn yellow.
- She repeated the experiment at 20 °C, 25 °C and 30 °C, using fresh indicator and different woodlice each time.

Temperature / °C	Time taken for indicator to turn yellow / min
15	20
20	14
25	10
30	7

(a) State the independent variable in Amara's investigation. [1]

(b) Give two variables that Amara should keep the same. [2]

(c) Explain why the indicator turned yellow more quickly at higher temperatures. [3]

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14 · Part 3

Paper 1 & 2 Respiration [6]

The table shows the energy released when 1 g of glucose is respired in two different ways.

Type of respiration	Energy released / kJ per g of glucose
aerobic respiration	16.0
anaerobic respiration in muscle cells	1.2

(a) Calculate the energy released by anaerobic respiration as a percentage of the energy released by aerobic respiration.

Use the formula: percentage = (energy released by anaerobic respiration ÷ energy released by aerobic respiration) × 100 [2]

(b) Explain why anaerobic respiration in muscle cells releases less energy from each gram of glucose than aerobic respiration. [2]

(c) Some types of bacteria can only respire anaerobically. Other types respire aerobically.

Suggest why bacteria that can only respire anaerobically grow more slowly than bacteria that respire aerobically, when both are given the same mass of glucose. [2]

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15 · Part 3

Paper 1 & 2 | Gas exchange | [6]

The human lungs are found in the thorax. Air enters the lungs through the trachea.

(a) Describe the path taken by air from the trachea to the alveoli. [3]

(b) State the name of the bones that surround and protect the lungs. [1]

(c) Describe the position of the pleural membranes in the thorax. [2]

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16 · Part 3

Paper 1 & 2 Gas exchange [6]

Leila made a model of the human thorax.

- She cut the bottom off a clear plastic bottle and tied a thin rubber sheet tightly across the open end.
- She pushed a Y-shaped glass tube through a hole in the bottle cap, so that the single end of the tube was open to the air outside the bottle.
- Inside the bottle, she attached a small balloon to the end of each of the two branches of the tube.
- The rubber sheet represents the diaphragm.

When Leila pulled the centre of the rubber sheet downwards, the balloons inflated.

(a) State what the balloons represent. [1]

(b) Explain why the balloons inflated when Leila pulled the rubber sheet downwards. [3]

(c) Give two ways in which Leila's model is different from the human thorax. [2]

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17 · Part 3

Paper 1 & 2 | Gas exchange | [6]

Oxygen and carbon dioxide are exchanged between the air in the alveoli and the blood.

(a) Explain how the alveoli are adapted for efficient gas exchange by diffusion. [4]

(b) The total surface area of the alveoli in the lungs of an adult is about 70 m^2. The surface area of the skin of the same adult is about 1.75 m^2.

Calculate how many times larger the surface area of the alveoli is than the surface area of the skin. [2]

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18 · Part 3

Paper 1 & 2 Gas exchange [7]

Scientists recorded the number of deaths from coronary heart disease in a large group of men over several years. The men were grouped by the number of cigarettes they smoked each day.

Number of cigarettes smoked each day	Deaths from coronary heart disease per 100 000 men per year
0	160
1 to 14	290
15 to 24	380
25 or more	480

(a) Describe the relationship between the number of cigarettes smoked each day and the death rate from coronary heart disease. [2]

(b) Calculate the percentage increase in the death rate from coronary heart disease for men who smoked 25 or more cigarettes a day, compared with men who did not smoke.

Use the formula: percentage increase = ((new value − original value) ÷ original value) × 100 [2]

(c) Explain how smoking can lead to coronary heart disease. [3]

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19 · Part 3

Paper 1 & 2 | Gas exchange | [6]

The airways leading to the lungs are lined with cells. Some of these cells make mucus, which traps dust and micro-organisms. Other cells have cilia, which beat to move the mucus up and away from the lungs.

(a) Chemicals in tobacco smoke damage the cilia.

Explain why smokers often have a cough. [3]

(b) Some long-term smokers develop emphysema. In emphysema, the walls between the alveoli break down, so the lungs contain fewer, larger air spaces.

Explain why a person with emphysema becomes breathless during exercise. [3]

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20 · Part 3

Paper 1 & 2 | Gas exchange | [8]

Tomas investigated the effect of exercise on his breathing rate.

- He sat still for five minutes and then counted the number of breaths he took in one minute.
- He did step-ups for three minutes and then immediately counted the number of breaths he took in one minute.
- He did this three times, resting between trials until his breathing rate returned to normal.

Trial	Breathing rate at rest / breaths per minute	Breathing rate after exercise / breaths per minute
1	14	26
2	15	29
3	13	26

(a) Calculate the mean breathing rate after exercise.

Use the formula: mean = sum of the values ÷ number of values [2]

(b) Explain why Tomas's breathing rate increased during exercise. [3]

(c) Suggest why Tomas sat still for five minutes before he counted his breathing rate at rest. [1]

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(d) Tomas also wanted to show that the air he breathes out contains more carbon dioxide than the air he breathes in. Limewater turns from clear to cloudy when carbon dioxide is bubbled through it.

Describe how Tomas could use limewater to show this. [2]

21 · Part 3 Paper 2 only Gas exchange [6]

Leaves of flowering plants exchange carbon dioxide and oxygen with the air around them.

(a) Explain how the structure of a leaf is adapted for gas exchange. [4]

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(b) Explain why the concentration gradient of carbon dioxide between the air outside a leaf and the inside of the leaf is steeper in bright light than in dim light. [2]

22 · Part 3

Paper 2 only

Gas exchange

[6]

Chen counted the stomata on the upper and lower surfaces of the leaves of two plants. Plant R grows on land. Plant S is a pond plant whose leaves float flat on the surface of the water, with the lower surface touching the water.

Plant	Number of stomata per mm^2 on upper surface	Number of stomata per mm^2 on lower surface
R	30	270
S	420	0

(a) Describe the role of stomata in gas exchange. [2]

(b) Calculate the percentage of the stomata of plant R that are on the lower surface of the leaf.

Use the formula: percentage = (number of stomata on lower surface ÷ total number of stomata on both surfaces) × 100 [2]

(c) Suggest why the leaves of plant S have stomata only on the upper surface. [2]

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23 · Part 3

Paper 2 only

Gas exchange

[7]

Rosa placed a potted plant in a sealed, transparent chamber. She measured the net rate of carbon dioxide uptake by the plant at different light intensities. The temperature was kept the same throughout. A negative value means that the plant released carbon dioxide.

Light intensity / arbitrary units	Net rate of carbon dioxide uptake / arbitrary units
0	−4
10	0
20	4
30	8
40	10
50	10

(a) State the light intensity at which there was no net exchange of carbon dioxide. [1]

(b) Explain why there was no net exchange of carbon dioxide at this light intensity. [2]

(c) Explain the result at a light intensity of 0 arbitrary units. [2]

(d) Assume that the rate of respiration of the plant was the same at all light intensities.

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Calculate the rate at which carbon dioxide was used in photosynthesis at a light intensity of 40 arbitrary units.

Use the formula: rate of carbon dioxide use in photosynthesis = net rate of carbon dioxide uptake + rate of carbon dioxide release by respiration [2]

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24 · Part 3

Paper 2 only

Gas exchange

[7]

The colour of hydrogen-carbonate indicator depends on the concentration of carbon dioxide. From the highest to the lowest carbon dioxide concentration, the colours are: yellow, orange, red, magenta, purple. The indicator is red when it is in equilibrium with normal air.

Femi investigated the effect of light on net gas exchange in leaves.

- He put 5 cm^3 of red hydrogen-carbonate indicator into each of six boiling tubes.
- He placed one leaf of the same plant, of similar size, in each of five tubes, held above the indicator on a gauze platform. The sixth tube had no leaf.
- He sealed each tube with a bung.
- He placed the five tubes with leaves at different distances from a lamp. He placed the tube with no leaf 10 cm from the lamp.
- After two hours, he recorded the colour of the indicator in each tube.

Tube	Distance from lamp / cm	Colour of indicator after two hours
leaf	10	purple
leaf	20	magenta
leaf	30	red
leaf	40	orange
leaf	50	orange
no leaf	10	red

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(a) Give the dependent variable in Femi's investigation. [1]

(b) Explain the result for the leaf 10 cm from the lamp. [3]

(c) The lamp also gives out heat. Explain why this could affect Femi's results, and suggest how he could reduce this effect. [2]

(d) Suggest why Femi included a tube with no leaf. [1]

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25 · Part 3

Paper 2 only

Gas exchange

[6]

In a small pond there are water plants and animals. Scientists measured the concentration of dissolved oxygen in the pond water every three hours for one day. The sun rose at 06:00 and set at 18:00.

Time of day	Concentration of dissolved oxygen / mg per dm^3
06:00	4.2
09:00	5.4
12:00	7.2
15:00	8.6
18:00	9.4
21:00	7.6
00:00	6.0
03:00	4.9
06:00 (next day)	4.2

(a) Describe how the concentration of dissolved oxygen changed over the 24 hours. [2]

(b) Explain the changes in the concentration of dissolved oxygen over the 24 hours. [4]

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